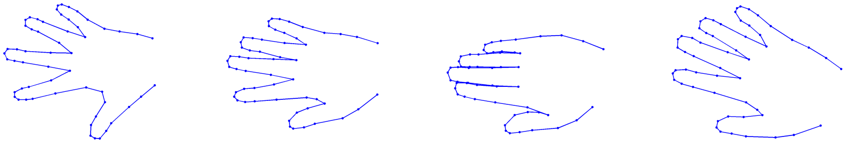
# Matrix Theory: Lecture 10

Carl Olsson

2025-11-03

# Low Rank Models



Four (out of 40) images of a deformable model with 56 tracked point.

- Point movements are not independent
- No explicit model
- Extract linear model from observed data

# Low Rank Models

$(x_{ij}, y_{ij})$ - coordinates of point $i$ on hand $j$.
Measurement matrix

$$M = \begin{pmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1n} \\ y_{11} & y_{12} & y_{13} & \cdots & y_{1n} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2n} \\ y_{21} & y_{22} & y_{23} & \cdots & y_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & x_{m3} & \cdots & x_{mn} \\ y_{m1} & y_{m2} & y_{m3} & \cdots & y_{mn} \end{pmatrix}.$$

Column $=$ point in $\mathbb{R}^{2m}$.

Can we find a (linear) function of $r$ parameters
that gives us all columns/hand-shapes?

# Low Rank Models

- If $f$ is linear then it has a matrix representation $B$.
- If $c_i \in \mathbb{R}^r$ are the parameters that give $M_i \in \mathbb{R}^r$

$$M_i = f(c_i) = Bc_i$$

- Stacking all $M_i$ in $M$ gives.

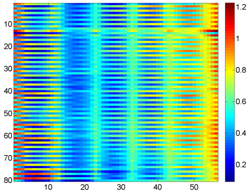
$$M = \begin{bmatrix} M_1 & M_2 & \dots & M_n \end{bmatrix} = B \begin{bmatrix} c_1 & c_2 & \dots & c_n \end{bmatrix} = BC$$

  where $B - 2m \times r$, $C - r \times n$.
- If $f$ exists we must have $\text{rank}(M) = r$.

# Low Rank Models

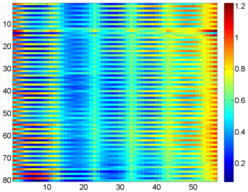
Can we approximate M with a low rank matrix?

Original $M$



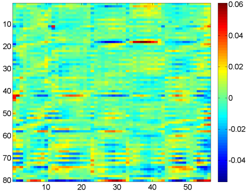
$\text{rank}(M) = 56$

Approximation



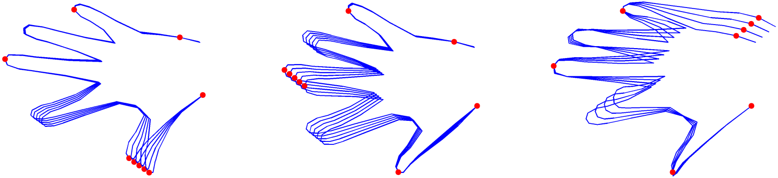
$\text{rank}(X) = 5$

Difference



Approximation computed with SV-decomposition.

When $f$ is known we can generate new shapes by specifying $c$ and computing $f(c)$:



# Singular Values

**Theorem 13.1**

If $A$ is an arbitrary matrix (not necessarily square) then the matrix $B = A^H A$ is a Hermitian matrix with nonnegative eigenvalues.
In addition $\operatorname{Ker} A = \operatorname{Ker} B$.

Def. If $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_r > 0$ the singular values are $\sigma_i = \sqrt{\lambda_i}$.

# The Singular Value Decomposition

## Theorem 13.2

(**SV-decomposition**) Every $m \times n$ matrix $A$ can be decomposed as

$$A = USV^H,$$

where $U$ and $V$ are unitary matrices (of size $m$ and $n$ respectively), and $S$ is a $m \times n$ block matrix of form

$$S = \left(\begin{array}{cccc|c} \sigma_1 & 0 & \cdots & 0 & \\ 0 & \sigma_2 & \cdots & 0 & \\ \ldots & \ldots & \ldots & \ldots & O \\ 0 & 0 & \cdots & \sigma_r & \\ \hline & & O & & O \end{array}\right)$$

Here the non-zero block is a diagonal $r \times r$ matrix having all singular values $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r$ on the diagonal and zero blocks have sizes $m - r \times r, r \times n - r, m - r \times n - r$.

# Singular Value Decomposition

$V_i$ - orthonormal eigenvect. of $B = A^H A$, that is, $BV_i = \begin{cases} \sigma_i^2 V_i & i \leq r \\ 0 & i > r \end{cases}$.

$U_i := \frac{1}{\sigma_i} A V_i \Leftrightarrow U_i \sigma_i = A V_i \Leftrightarrow$

$$\begin{pmatrix} U_1\sigma_1 & \dots & U_r\sigma_r \end{pmatrix} = A \begin{pmatrix} V_1 & \dots & V_r \end{pmatrix} \Leftrightarrow$$

$$\underbrace{\begin{pmatrix} U_1 & \dots & U_r \end{pmatrix}}_{m\times r} \underbrace{\begin{pmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \sigma_r \end{pmatrix}}_{:=\Sigma\ (r\times r)} = \underbrace{A}_{m\times n} \underbrace{\begin{pmatrix} V_1 & \dots & V_r \end{pmatrix}}_{n\times r}$$

# Singular Value Decomposition

$V_i$ - orthonormal eigenvect. of $B = A^H A$, that is, $BV_i = \begin{cases} \sigma_i^2 V_i & i \leq r \\ 0 & i > r \end{cases}.$

$U_i := \frac{1}{\sigma_i} A V_i \Leftrightarrow U_i \sigma_i = A V_i \Leftrightarrow$

$$\begin{pmatrix} U_1\sigma_1 & \dots & U_r\sigma_r \end{pmatrix} = A \begin{pmatrix} V_1 & \dots & V_r \end{pmatrix} \Leftrightarrow$$

$$\underbrace{\begin{pmatrix} U_1 & \dots & U_r & \color{red}{U_{r+1}} & \color{red}{\dots} & \color{red}{U_m} \end{pmatrix}}_{m \times m} \underbrace{\begin{pmatrix} \Sigma \\ \color{red}{0} \end{pmatrix}}_{m \times r} = \underbrace{A}_{m \times n} \underbrace{\begin{pmatrix} V_1 & \dots & V_r \end{pmatrix}}_{n \times r}$$

# Singular Value Decomposition

$V_i$ - orthonormal eigenvect. of $B = A^H A$, that is, $BV_i = \begin{cases} \sigma_i^2 V_i & i \leq r \\ 0 & i > r \end{cases}$.

$U_i := \frac{1}{\sigma_i} A V_i \Leftrightarrow U_i \sigma_i = A V_i \Leftrightarrow$

$$\begin{pmatrix} U_1\sigma_1 & \dots & U_r\sigma_r \end{pmatrix} = A \begin{pmatrix} V_1 & \dots & V_r \end{pmatrix} \Leftrightarrow$$

$$\underbrace{U}_{m\times m} \underbrace{\begin{pmatrix} \Sigma \\ 0 \end{pmatrix}}_{m\times r} = \underbrace{A}_{m\times n} \underbrace{\begin{pmatrix} V_1 & \dots & V_r \end{pmatrix}}_{n\times r}$$

# Singular Value Decomposition

$V_i$ - orthonormal eigenvect. of $B = A^H A$, that is, $BV_i = \begin{cases} \sigma_i^2 V_i & i \leq r \\ 0 & i > r \end{cases}$.

$U_i := \frac{1}{\sigma_i} A V_i \Leftrightarrow U_i \sigma_i = A V_i \Leftrightarrow$

$$\begin{pmatrix} U_1\sigma_1 & \ldots & U_r\sigma_r \end{pmatrix} = A \begin{pmatrix} V_1 & \ldots & V_r \end{pmatrix} \Leftrightarrow$$

$$\underbrace{U}_{m\times m} \underbrace{\begin{pmatrix} \Sigma & \color{red}{0} \\ 0 & \color{red}{0} \end{pmatrix}}_{m\times n} = \underbrace{A}_{m\times n} \underbrace{\begin{pmatrix} V_1 & \ldots & V_r & \color{red}{V_{r+1}} & \color{red}{\ldots} & \color{red}{V_n} \end{pmatrix}}_{n\times n} \Leftrightarrow$$

$$US = AV \Leftrightarrow USV^H = A.$$

# Sigular Value Decomposition (SVD)

**Theorem 13.3**

Let $A$ be an arbitrary $m \times n$ matrix.

1. $r(A) = r(A^H A) = r(AA^H) = r(A^H) =$ number $r$ of singular values.
2. If $A = USV^H$ with unitary $U, V$ and a matrix $S$ with only $k$ positive numbers on the main diagonal $s_1 \geq s_2 \geq \ldots \geq s_k > 0$ then this an SVD, and $s_i = \sigma_i$.
3. The matrix $A^H$ has the same singular values $\sigma_1, \ldots, \sigma_r$ as $A$ and if $A = USV^H$ is an SVD for $A$ then $A^H = VS^T U^H$ is an SVD for $A^H$.
4. The columns $U_1, \ldots, U_r$ in any SVD of $A$ form an orthonormal basis of $\operatorname{Im} A$ and $U_{r+1}, \ldots, U_m$ form an orthonormal basis of $\operatorname{Ker} A^H$.
5. The columns $V_1, \ldots V_r$ form an orthonormal basis of $\operatorname{Im} A^H$ and $V_{r+1}, \ldots, V_n$ form an orthonormal basis of $\operatorname{Ker} A$.
6. $A = \sum_{i=1}^{r} \sigma_i U_i V_i^H$ is a sum of $r$ matrices of rank 1.

# Sigular Value Decomposition (SVD)

**Theorem 13.4**

Let $A$ be a square matrix and $\sigma_1, \ldots, \sigma_r$ be its singular values. Then

1. $\|A\|_2 = \sigma_1$.
2. If $A$ is invertible then $A^{-1}$ has singular values equal to $\frac{1}{\sigma_r}, \ldots, \frac{1}{\sigma_1}$. (Note that $r$ is equal to the size of the matrix in this case).
3. If $A$ is invertible then its condition number is equal to $\kappa(A) = \frac{\sigma_1}{\sigma_r}$.
4. $\|A\|_F = \sqrt{\sigma_1^2 + \cdots + \sigma_r^2}$ for the Frobenius norm.

# Ex 13.6

Suppose that $A$ has rank $r$ and $\sigma_1, ..., \sigma_r$ are its singular values. Suppose that $k < r$ and consider all matrices X of the same size and rank $k$. Show that

$$\sigma_{k+1} = \min_X \|A - X\|_2$$

where the minimum is taken over all such matrices $X$.